

EcoTune: Edge-Cloud Collaborative Model Adaptation for Budget-Constrained On-Device SLM Personalization

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Abstract

The rapid growth of web content has spurred the widespread adoption of on-device AI assistants powered by large language models (LLMs). However, deploying and personalizing these assistants in real-world environments remains challenging due to limited annotation budgets and scarce on-device fine-tuning resources. Existing edge-cloud collaboration frameworks typically rely on costly cloud-based supervision or perform full-layer fine-tuning, leading to inefficiencies in both computation and adaptation. To address these limitations, we propose EcoTune, a budget-constrained framework for efficient edge-cloud collaborative adaptation. EcoTune jointly optimizes representative data selection for cloud annotation and selective on-device model adaptation within a unified closed-loop process. Specifically, it employs a multi-armed bandit-based strategy to identify high-value user interactions for cloud supervision and a layer importance-driven adaptation mechanism to update only critical components of the small language model (SLM). This coordinated optimization enables dynamic, resource-efficient personalization under stringent annotation and tuning budgets. Experiments on real-world testbeds demonstrate that EcoTune achieves up to 20%-60% reduction in annotation costs and significantly lowers fine-tuning memory consumption compared to state-of-the-art baselines, providing a practical and scalable solution for personalized on-device LLMs.

CCS Concepts

• **Computing methodologies** → **Machine learning: Cooperation and coordination.**

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Keywords

Large Language Model, finetuning, Edge-Cloud Collaboration, Small Language Model

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1 Introduction

As web content continues to grow at an exponential rate, on-device AI assistants have become indispensable tools for helping users navigate an increasingly complex and information-rich online environment. Widely adopted systems such as Google Assistant, Microsoft Cortana, and Amazon Alexa [22] have demonstrated effectiveness in a range of everyday tasks, including question answering, medical consultations, and creative writing. With the rapid advancement of large language models (LLMs) and their emergent capabilities in commonsense reasoning, task planning, and in-context learning, the development of LLM-powered personal assistants has emerged as both a critical research frontier and a key direction for real-world applications [29]. Notable efforts in this domain include WebWalker [47] and Apple Intelligence [1], both of which support intelligent, context-aware user interactions. Given the high latency and substantial cost of cloud-based LLM services [29], recent works focus on edge-cloud collaborative frameworks that deploy personalized small language models (SLMs) (e.g., LLaMA-3 series [18], Qwen2.5 series [51]) on device, while leveraging powerful cloud LLMs for supervision. In this paradigm, cloud LLMs act as black-box annotators, generating pseudo-labels from on-device interaction data (e.g., queries, dialogues) to personalize local SLMs [28, 31, 34, 48].

Despite their advantages, this paradigm faces critical challenges under real-world deployment for domain-specific tasks adaptation. First, label scarcity remains a fundamental obstacle. Unlike offline

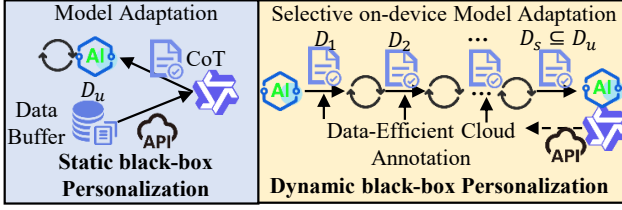


Figure 1: Static Method vs. EcoTune

datasets used in cloud-based training, which are often curated using gradient-based selection methods that rely on ground-truth labels [5, 12, 19, 49, 54], on-device assistants continuously accumulate large volumes of raw interaction data (e.g., queries, dialogues, web contents) that inherently lack access to ground truth. Users rarely provide explicit annotations or correct responses due to limited incentives and expertise [4, 55, 57]. While cloud LLMs can serve as black-box annotators, repeated API calls incur significant monetary and latency costs [15, 42]. Thus, it becomes crucial to intelligently select representative data samples for cloud annotation, focusing limited resources on high-value instances. Second, on-device fine-tuning resources are severely limited. While Parameter-Efficient Fine-Tuning (PEFT) methods [20, 21, 23, 26] like Low-Rank Adaptation (LoRA) [21] reduce trainable parameters, applying them across all layers still introduces substantial memory and storage overhead. For instance, LoRA fine-tuning of LLaMA2-7B requires up to 24GB memory [53], which remains prohibitive for edge devices. Thus, it is equally important to selectively adapt only the most critical model components [9, 27, 32, 53], minimizing resource usage per update. Additionally, existing approaches address these challenges in isolation but fail to exploit the synergy between them, making them unsuitable for dynamic, long-term personalization.

To tackle these challenges, we propose EcoTune, a budget-aware framework for edge-cloud collaborative model adaptation that co-optimizes representative data selection and selective model adaptation on edge devices, enabling efficient personalization under stringent annotation and tuning budgets. As illustrated in Fig. 1(left), traditional static black-box personalization methods [28, 34] adapt SLMs using all pseudo-labeled domain-specific data generated by cloud LLM, regardless of data quality or model learning progress. In contrast, EcoTune integrates data-efficient cloud annotation and selective on-device model adaptation into a unified, iterative workflow. It periodically selecting representative samples for LLM annotation while fine-tuning only critical components of the SLM. This design allows EcoTune to capture the dynamics of the training process and to achieve efficient, long-term personalization of SLMs within constrained computational resources.

Specifically, the framework of EcoTune follows an iterative refinement paradigm that jointly optimizes representative data for cloud annotation and selective on-device model adaptation in a closed loop. It first identifies high-value interactions for black-box supervision using a multi-armed bandit strategy. Then EcoTune selectively updates only the most critical components of the model based on the pseudo labels via layer importance metric. The updated model will guide the next round to capture the dynamic of

training process. This coordinated process enables substantial improvements in downstream performance while markedly reducing annotation and fine-tuning costs, making on-device adaptation both practical and scalable. We implement EcoTune based on the top-tier LLaMA2 series models and evaluate it on a real-world testbed. Experimental results show that EcoTune outperforms all baselines, reducing annotation costs by up to 20%–60% while maintaining comparable performance across multiple benchmarks.

Our work makes the following contributions:

- We present EcoTune, a budget-constrained framework for edge–cloud collaborative model adaptation that jointly optimizes data-efficient cloud annotation and selective on-device model adaptation into a closed loop, enabling efficient personalization under stringent annotation and tuning budgets.
- We propose a multi-armed bandit-based data acquisition strategy to reduce reliance on costly cloud LLM annotations, coupled with a selective model adaptation mechanism that dynamically identifies critical layers using a training-aware importance metric. Together, these components enable efficient supervision and adaptive refinement of on-device models while capturing the evolving nature of the learning process.
- We implement EcoTune on a real-world testbed. Extensive experiments across diverse benchmarks demonstrate that EcoTune reduces annotation costs by up to 20%–60%, achieves performance on par with state-of-the-art baselines, and significantly lowers memory consumption during fine-tuning.

2 Related Works

2.1 Data Selection for LLMs

The curation of training data is critical for producing highly performance LLMs. In data center settings, conventional data selection often rely on gradient-based approaches, which typically require access to ground-truth labels [5, 12, 19, 49, 54]. For instance, DataRater [5] estimates the value of training data by meta-learning approach to enhance the efficiency of LLM training. Similarly, [12, 19] train a data scorer on a proxy dataset by analyzing the gradient direction of each sample to assess its quality. However, these methods require computing the gradient impact of each data point and rely on access to ground-truth labels, both of which are prohibitively expensive for resource-constrained edge devices, where high-quality annotated data is typically scarce. Some recent approaches explore heuristic-based methods to approximate the impact of selected data on model performance [8, 13, 39, 46]. For instance, [13] and [8] propose one-shot data selection frameworks based on clustering techniques. STAR [55] introduces a dynamic uncertainty measurement framework for data-efficient fine-tuning. Ask-LLM [39] and QuRating [46] leverage the reasoning capabilities of LLMs to directly assess the quality of training examples. Although these methods do not rely on ground-truth labels, they are inherently static and fail to capture how model preferences over data evolve during the training process.

2.2 Layer-Wise Importance Finetuning

Numerous studies have revealed the presence of layer redundancy in pre-trained language models [9, 17, 32, 36, 53], suggesting that

not all layers contribute equally to downstream task performance. This motivates selective fine-tuning strategies that adapt only the most critical layers, reducing computational and memory overhead. However, identifying which layers to fine-tune remains a key challenge. LISA [36] randomly selects a subset of transformer layers for optimization, achieving faster convergence and improved performance. Liu et al. [32] identify critical layers by analyzing significant shifts in representation space across training stages. OWLore [27] assigns higher sampling probabilities to layers with more neuron-level outliers, enabling selective fine-tuning of only a few layers. Nevertheless, most existing methods rely on pre-defined, static criteria, such as weight norm [36] or outlier statistics [27], determined prior to training. These static metrics fail to capture the evolving functional roles of layers during adaptation, limiting their effectiveness in dynamic personalization scenarios. In contrast, EcoTune introduces a dynamic layer importance metric that adapts throughout training, enabling effective model updates.

2.3 Edge-Cloud Collaborative LLM Fine-tuning

Personalized LLMs must support domain-specific tasks, posing significant challenges on computational resources and memory footprint owing to the vast number of parameters in LLMs [6, 7, 24, 52]. Leveraging PEFT methods [20, 21, 23, 26] offers a compelling approach to fine-tune selective LLMs without necessitating the adjustment of the entire model’s parameters. LoRA [21] delivers state-of-the-art results among various PEFT strategies using two low-rank matrices. To obtain high-quality supervision for SLM personalization, black-box adaptation [28, 30, 31, 34, 48] has shown promising results in fine-tuning small models on the prompt-response pairs generated by LLM APIs, without requiring local LLM deployment. [34] finetuned a student model on the chain of thought (CoT) outputs generated by a larger teacher model to explore the transfer of such reasoning capabilities to smaller models via knowledge distillation. [30] proposed Mix Distillation, a simple yet effective strategy that balances reasoning complexity by combining long and short CoT examples or reasoning from both larger and smaller models. However, these approaches typically treat all training data as equally important, making them ill-suited for on-device personalization, where data is unlabeled and user behavior evolves dynamically over time. Moreover, repeatedly querying cloud LLMs to generate chain-of-thought rationales incurs substantial API costs, when dealing with large volumes of user interaction data.

3 Motivation

3.1 Under-Utilization of On-Device Data

On-device fine-tuning enables the personalization of models by leveraging locally collected user interactions, allowing for improved relevance and privacy preservation. However, these interactions are often unlabeled and of heterogeneous quality, making reliable supervision both costly and inefficient when relying on cloud LLM queries or human annotation. Traditional black-box approaches to personalization typically treat all interaction data equally, leading to wasteful allocation of limited annotation budgets on redundant or low-value samples that contribute little to model improvement.

To better understand this problem, we conducted experiments to evaluate the impact of annotation data obtained via cloud-based

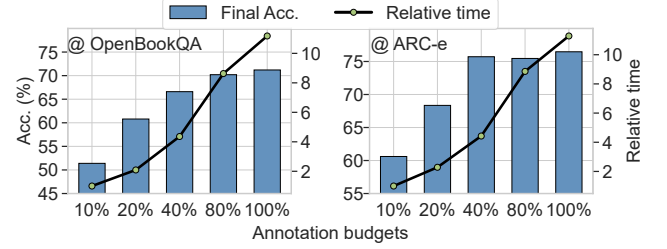


Figure 2: Zero-shot accuracy and normalized training time of LLaMA2-7B with LoRA on OpenBookQA and ARC-e under varying annotation budgets.

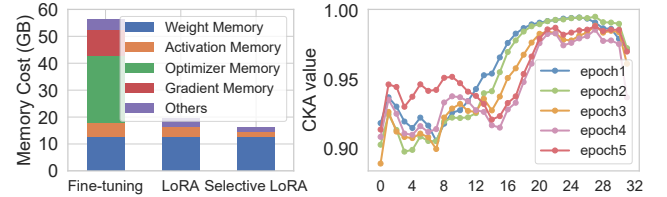


Figure 3: Left: Memory breakdown of various methods using LLaMa2-7B on OpenBookQA dataset. Right: Average CKA values of different layers across epochs in training.

LLM queries. Specifically, we fine-tune the LLaMA2-7B model using LoRA on the OpenBookQA [35] and ARC-e [10] benchmarks (see §5). As illustrated in Fig. 2, relying on only partial annotation data results in a 15.86%–19.8% drop in final inference accuracy, underscoring that not all data contributes equally to model improvement. Additionally, each additional 20% of annotated data leads to a 1.2×–2.42× increase in training time, exacerbating resource bottlenecks on edge devices. These findings emphasize the importance of intelligent data selection to optimize personalization efficiency. Rather than annotating uniformly, we should prioritize samples that maximize learning gain per annotation budget.

3.2 Static Model Adaptation

Static black-box personalization typically involves either uniform updates across all transformer layers [21, 23] or the selection of a fixed set of critical layers determined prior to training [9, 27, 32, 36]. As shown in Fig. 3(left), while these approaches reduce trainable parameters compared to full fine-tuning, their uniform or one-shot nature proves suboptimal in practice, particularly under dynamic training conditions.

To investigate this limitation, we analyze representation dynamics by computing the Centered Kernel Alignment (CKA) [25] similarity across layers at each training epoch, using different annotated data subsets. CKA is a widely adopted metric for comparing neural network representations, capturing semantic similarities between layer activations [17, 27, 32]. A lower CKA value indicates a greater divergence in representations, reflecting more significant functional shifts. In Fig. 3(right), we plot the CKA curves for different SLM layers across training epochs. We observe substantial representation shifts over time, with varying patterns across layers. This

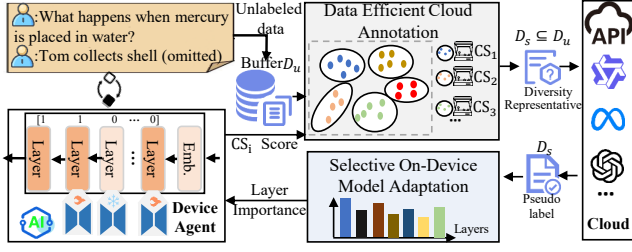


Figure 4: Overall Framework of EcoTune.

demonstrates that the functional roles of layers evolve dynamically during adaptation, rendering static selection strategies ineffective. These findings highlight the need for adaptive layer updating mechanisms that respond to both evolving model representations and heterogeneous data distributions.

4 EcoTune Overview

To enable SLM actively adapts to the evolving user interactions under cloud annotation budget constraints, we introduce EcoTune, a budget-constrained collaborative model adaptation framework that jointly optimizes data-efficient cloud annotation and selective on-device model adaptation, as shown in Fig. 4. EcoTune adopts an iterative refinement framework that seamlessly integrates these two module into a closed loop. In each round, it first identifies high-value user interactions for black-box supervision through a multi-armed bandit strategy, which ensures the representativeness and diversity of training data. Guided by the resulting pseudo-labels, EcoTune then selectively updates only the most critical components of the model, as determined by a layer importance metric across the training process. The refined model subsequently informs the next iteration, enabling the system to adapt dynamically to evolving training signals and data distributions.

4.1 Budget-Constrained SLM Personalization

LLMs inherently generate sentences in a free-form and in an autoregressive manner [44]. This process entails the sequential prediction of the probability distribution for the subsequent token in a sentence. Let x represent the input prompt, and $\hat{y}$ denote the sentence generated by the LLM, comprising N tokens in total. For any given LLM, the probability of producing a specific token z_i as the i -th element in the sentence can be mathematically expressed as $p(z_i | \hat{y}_{<i}, x)$, where $1 \leq i \leq N$. Here, $\hat{y}_{<i}$ symbolizes the sequence of previously generated tokens $\{z_1, z_2, \dots, z_{i-1}\}$.

Our goal is to fine-tune an on-device SLM θ to better support domain-specific tasks using a replay buffer D_u , which continuously accumulates user interactions with the assistant depending on system load [33]. In practice, fine-tuning can be performed opportunistically, for instance, only during periods of low service demand when spare computational resources (e.g., FLOPs) are available. To maximize the fine-tuning benefit under limited black-box supervision, it is necessary to select a representative subset $D_s \subseteq D_u$ for cloud LLM annotation achieving pseudo-label D'_s . This subset should maintain high data quality, capture diverse interaction patterns, and provide maximal training utility. Furthermore, instead of

updating all parameters, we selectively adapt only the most critical layers $L' \subseteq L$ of θ to reduce memory consumption while preserving strong generalization performance on a downstream evaluation dataset D_{test} with a loss function $\mathcal{L}$. The problem can thus be formulated as a budget-constrained optimization task:

$$\begin{aligned} \min_{\{D_s, L' \subseteq D_u, L\}} \quad & \mathcal{L}(x_i, y_i) \sim D_{test}(\hat{y}_i | x_i, y_i) \\ \text{s.t.} \quad & P(D_u) \leq \mathcal{B} \\ & L' = \text{Select}(L | D'_s) \end{aligned} \quad (1)$$

where $P(\cdot)$ denotes the API cost of querying a cloud LLM for an annotation, computed based on the number of input and generated tokens, under a total budget constraint $\mathcal{B}$; L is the total number of layers in the SLM, and L' represents the subset of critical layers selected during training. The above budget-constrained problem is non-trivial due to the large parameter space of SLMs and the limited computational resources available on edge devices. In addition, the non-differentiable nature of data selection and selective model adaptation further complicates the optimization. In the next section, we present a principled closed-loop iterative refinement paradigm that systematically addresses these challenges.

4.2 Data-Efficient Cloud Annotation.

As emphasized in the EcoTune framework, effective data selection requires identifying a representative and diverse subset that captures edge cases in the original dataset, forming the foundation for SLM fine-tuning. Instead of directly selecting in the discrete data space, it is often more efficient and practical to operate in the feature space. Therefore, each data sample x_i is first mapped into a high-dimensional representation through a feature encoder E , resulting in $f_i = E(x_i)$, which is subsequently normalized. We adopt all-MiniLM-L6-v2 [37] as the embedding model. Despite its compact size, it achieves strong representational performance while maintaining high efficiency on edge devices. Additional implementation details are provided in Appendix A.1. The resulting feature set of the dataset is subsequently clustered into k cluster $\mathcal{K} = \{C_1, C_2, \dots, C_K\}$ to reveal the underlying structures and group similar instances, enabling a deeper understanding of the data distribution. Therefore, the cluster is denoted as:

$$C_i = \{f_j\}_{j=1}^{|C_i|}, 1 \leq i \leq K \quad (2)$$

Given that raw interaction data (e.g., queries, dialogues, web content) inherently lack ground-truth supervision, we leverage maximum-entropy-based uncertainty criteria to evaluate sample importance, thereby circumventing the expensive gradient computations required by SLM [5, 12, 19, 49]. Specifically, the maximum entropy (ME) of a generated response s given an input x is defined as:

$$\text{ME}(\hat{y}, x) = - \sum_{i=1}^N \sum_{j=1}^V p(v_{ij} | \hat{y}_{<i}, x) \log p(v_{ij} | \hat{y}_{<i}, x) \quad (3)$$

where $p(v_{ij} | \hat{y}_{<i}, x)$ denotes the probability of the j -th token in the vocabulary at the i -th position, and V is the vocabulary size. This metric quantitatively captures model uncertainty by computing the entropy over all possible token predictions. To adapt data selection to the evolving training dynamics, we cast it as an online learning problem and adopt a multi-armed bandit (MAB) framework. This

enables a principled trade-off between discovering potentially valuable and reusing clusters with historically high utility. As shown in Figure 4, each cluster corresponds to an arm of the bandit. In each iteration, the MAB policy selects a cluster based on its estimated value (e.g., average cluster score), from which a subset of samples is drawn for cloud annotation. This iterative process progressively refines the selection toward high-impact data.

Specifically, in each training iteration, we compute the cluster score based on the average maximum-entropy of the samples within the cluster and their sampling frequency. For each cluster, individual sample scores are first calculated, and high-scoring samples are selected for annotation and training. These samples then contribute to updating the cluster’s overall score, enabling the MAB algorithm to progressively refine the quality estimation of each cluster. At the same time, the algorithm ensures diversity by occasionally sampling from less-visited clusters. To strike a principled balance between exploration (i.e., data diversity) and exploitation (i.e., data quality), we adopt the Upper Confidence Bound (UCB) [2] strategy to evaluate each cluster score, defined as:

$$CS_i = \bar{M}E_i + \alpha \sqrt{\frac{2 \ln \left(\sum_j T(C_j) \right)}{T(C_i)}} \quad (4)$$

where the first term, $\bar{M}E_i$, denotes the average maximum-entropy score of cluster C_i , while the second term represents the exploration bonus. Here, $T(C_i)$ is the frequency of samples drawn from cluster C_i , and $\sum_j T(C_j)$ is the total number of samples selected across all clusters.

4.3 Selective On-Device Model Adaptation

In resource-constrained on-device adaptation, improving training efficiency and reducing memory overhead require selectively updating only the most influential model layers. However, layer importance cannot be determined in isolation. It depends on both the functional uniqueness of a layer’s representations and the sensitivity of the model’s outputs to updates in that layer.

Specifically, we first take a spectral perspective to capture representational uniqueness between layers. For the relationship with other layers, we employ Centered Kernel Alignment (CKA) [3, 32, 43, 58] to measure the similarity between a layer’s activations and those of other layers. Consider a batch of B input sequences, the l -th representations of a SLM with L layers is $X_l = [x_1, x_2, \dots, x_B]^T \in \mathbb{R}^{B \times d_l}$. Therefore, the linear CKA similarity between layers l_i and l_j is written as:

$$CKA(X_i, X_j) = \frac{\|X_j^T X_i\|_F^2}{\|X_i^T X_i\|_F \|X_j^T X_j\|_F} \quad (5)$$

where $\|\cdot\|_F$ denote the Frobenius norm. The average similarity of layer l_i to all other layers is:

$$R_i = \frac{1}{n-1} \sum_{j \neq i} CKA(X_i, X_j). \quad (6)$$

To quantify uniqueness, we normalize R_i and define the representational uniqueness score as:

$$U_i = 1 - \text{normalize}(R_i), \quad (7)$$

Layers with low similarity (i.e., high CKA distance) indicates lower redundancy and greater functional distinctiveness, suggesting a higher potential to contribute novel information during adaptation.

To assess a layer’s impact on model outputs, we use the magnitude of parameter updates in its LoRA adapters as a proxy for task sensitivity. Concretely, let $\mathcal{P}_i$ be the set of LoRA parameters in layer i and let $g_{p,t} = \|\nabla_{\theta_p} \ell_t\|_2^2$ be the squared gradient observed for parameter p at step t ; Aggregating over a recent update window $\mathcal{T}$, we compute the total gradient signal:

$$\begin{aligned} \tilde{G}_l &= \sum_{p \in \mathcal{P}_l} \sum_{t \in \mathcal{T}} g_{p,t} \\ G_l &= \frac{\tilde{G}_l}{|\mathcal{P}_l|} \end{aligned} \quad (8)$$

where G_l normalizes by the number of parameters to enable fair comparison across layers. By combining these two signals, we derive a composite and interpretable Layer Importance Metric (LIM) that identifies layers that are both representation unique and dynamically impactful, guiding our selective adaptation strategy:

$$M_l = \alpha \cdot G_l + (1 - \alpha) \cdot U_l \quad (9)$$

and we derive the layer sampling probability as:

$$p_l = \frac{M_l}{\sum_{j=1}^{j=L} M_j} \quad (10)$$

4.4 Overall EcoTune Algorithm

To capture the dynamics of the training process, we jointly optimizes data selection and selective model adaptation in a closed-loop iterative refinement paradigm. Initially, we generate embeddings for the on-device data and partition them into K clusters. In each iteration, we update the score of each cluster by equation (4). We then sample data from the highest-scoring cluster until the batch size reaches the predefined budget $|S|$ for cloud LLM annotation. Upon receiving pseudo-labels from the cloud annotator, we update the labeled dataset D_s and the unlabeled pool D_u . Subsequently, we identify the most critical layers of the SLM using equation (10) and fine-tune the model on D_s . This cycle repeats, refining both data and model adaptation, until the total annotation budget is exhausted. The complete workflow is summarized in Algorithm 1.

5 Experiments

5.1 Experimental Setup

Edge-Cloud Setting and Language Models. We use Qwen-2.5-Instruct-14B [51] as the cloud-based LLM for annotating unlabeled data. Details of the cloud model used for annotation can be found in Appendix A.2. For on-device deployment, we evaluate LLaMA-2-7B-chat-hf [18] chosen for their parameter scales being compatible with edge device constraints. All SLMs are deployed locally on an NVIDIA RTX 3090 GPU for simulation of on-device inference and fine-tuning. In the data-efficient cloud annotation process, the all-MiniLM-L6-v2 [37] is employed to generate embeddings for the input data, and the candidate pool D_u is clustered by k-means algorithm, where k is determined by silhouette coefficient method [38]. **Hyperparameter settings.** For PEFT training, we follow the configuration of the LLM-ADAPTER library [23]. We use a learning rate

Algorithm 1 EcoTune Algorithm

Input: Unlabeled training data $\mathcal{D}_u$, the SLM θ , number of iterations I , encoder model E , size of samples N for annotation each iteration

Output: Well-trained SLM θ .

```

1: Initialize  $C_s = \emptyset$ ,  $S = \emptyset$  and clustering on  $D_u$ 
2: for  $i = 0$  to  $I$  do
3:   Calculate the maximum entropy of samples by SLM  $\theta$  using
   (3), and set  $S = \emptyset$ 
4:   while  $|S| \leq N$  do
5:     Calculate each cluster score by using equation (4)
6:     Select samples from cluster  $C_{top}$  with the highest CS
7:     Update selected set  $S$ 
8:   end while
9:   Annotate  $S$  by cloud LLM
10:  Update  $D_s \leftarrow D_s \cup S$ ,  $D_u \leftarrow D_u \setminus S$ 
11:  Select top-K critical layers of SLM  $\theta$  by equation (10)
12:  Fine-tune SLM on  $D_s$  to get SLM  $\theta$ 
13: end for
14: return SLM  $\theta$  after finetuning

```

of 2×10^{-5} , LoRA rank of 16, warm-up ratio of 0.03, weight decay of 0.01, dropout ratio of 0.1, and train for 4 epochs. For selective on-device model adaptation, we conducted a sensitivity evaluation using different values of α . Additional experiments are provided in Appendix A.3 and for the main experiment, we set $\alpha = 0.6$. All experiments are implemented using the LLaMA-Factory [59].

Benchmarks. We evaluate the effectiveness of our budget-constrained framework on three domain-specific tasks using six open-source benchmarks. For each benchmark, we use the default dataset splits and treat the training set as unlabeled to simulate raw user interactions accumulated by an on-device SLM, mimicking the realistic scenario where labels are unavailable during data collection.

- **Arithmetic Reasoning.** GSM8K [11] contains approximately 8.5K diverse grade school math word problems. Each problem requires multi-step reasoning to derive the final answer.
- **Commonsense Reasoning.** OpenBookQA [35] is a four-way multiple-choice question-answering dataset designed to test commonsense knowledge. AI2 Reasoning Challenge (ARC) [10], a reasoning benchmark that includes a variety of grade-school level, multiple-choice science questions. This benchmark comprises two distinct subsets: ARC-Challenge (ARC-C) and ARC-Easy (ARC-E). CommonSenseQA [41] is a widely used dataset that places less emphasis on reasoning capabilities but requires a broader knowledge base.
- **Medical knowledge Reasoning.** MedConceptsQA [40] is an open-source benchmark for medical concepts question answering, including diagnoses, procedures, and medications. We focus on the ophthalmology subtask, which evaluates reasoning over eye-related conditions and treatments.

Baselines. We compare EcoTune with conventional finetuning and several methods based on data selection and layer-wise finetuning methods as follows:

- **STAR [55]:** This method employs a dynamic uncertainty estimation framework based on active learning [16, 48, 50]

for data-efficient fine-tuning. It selects samples with top-K prediction entropy to guide annotation.

- **Ask-LLM [39]:** This approach leverages the reasoning capabilities of LLMs to directly score data points for selection. It constructs a prompt asking whether a given instance is informative, and uses $P(\text{'yes'}|\text{prompt})$ as the sampling score. In our experiment, we use the Qwen-2.5-Instruct-14B model as LLM for scoring each data point.
- **DEFT-UCS [14]** This method employs unsupervised core-set selection to sample from both hard and easy instances with weighted sampling, balancing their contributions.
- **Flexora [45]:** This baseline treats layer selection as a hyperparameter optimization task, inferring the most effective layers based on observed update patterns. However, the need to jointly optimize both the selection policy and model parameters leads to a dual-optimization framework that is computationally expensive and substantially prolongs the training process.

Metric. We establish three metrics: 1) Accuracy: the ratio of correctly predicted outputs by the fine-tuned model under a fixed annotation budget. 2) C_{API} : the cost incurred during cloud-based annotation, computed by the length of input and output tokens. 3) C_{mem} : the peak memory usage on the edge device during inference and fine-tuning, measured in gigabytes (GB). We experiment with $D_s = \{10\%, 20\%, \dots, 80\%\}$ representing 10% to 80% of D_u

5.2 Main Results

Table 1 summarizes the accuracy of EcoTune and several baselines using LLaMA2-7B-chat-hf across three domain-specific tasks. As shown, EcoTune consistently outperforms all baselines on every benchmark, demonstrating the effectiveness of its joint optimization framework. For instance, EcoTune achieves up to a 3.62% improvement over STAR and a 4.37% improvement over DEFT-UCS. These gains highlight that, rather than relying on all available data, EcoTune intelligently selects representative and diverse samples at each iteration while selectively updating only the most critical layers of the model based on a dynamic layer importance metric. This joint optimization of data selection and model adaptation enables the model to progressively acquire complementary knowledge across training stages. In contrast, DEFT-UCS, the state-of-the-art method for unlabeled data selection, achieves the second-best performance but suffers from a one-shot selection strategy that fails to capture the dynamic nature of the training process, leading to accuracy drops of 0.99%, 1.87%, 1.07%, 0.13%, and 4.37% across several benchmarks (except MedConceptQA). Moreover, methods that treat data selection and selective model adaptation as independent objectives often exhibit suboptimal performance. For example, Flexora, which employs hyperparameter optimization to identify critical layers for fine-tuning, struggles to model the distributional diversity of heterogeneous on-device data, resulting in up to an 11.02% accuracy drop on complex arithmetic reasoning tasks. In addition, its gradient-based layer selection incurs higher training costs due to the dual learning process. Similarly, ASK-LLM requires constructing complex prompts to obtain quality scores from large LLMs, making it impractical for on-device SLM personalization because of its substantial API cost and latency overhead.

Methods	Arithmetic Reasoning	Commonsense Reasoning				Medical knowledge Reasoning
	GSM8K	OpenBookQA	ARC-C	ARC-E	CommonSenseQA	MedConceptsQA
STAR	37.83	71.44	61.14	75.89	74.33	40.42
Ask-LLM	39.57	72.12	60.04	73.44	70.77	42.7
DEFT-UCS	40.09	72.58	63.22	76.89	73.22	44.59
Flexora	30.02	68.93	55.45	70.27	71.03	38.59
Ours	41.08	74.45	64.29	77.02	77.59	43.48

Table 1: Zero-shot Performance of EcoTune and baselines on 6 benchmarks with all data. In each benchmark, the highest reasoning accuracy is highlighted in bold.

To evaluate the memory efficiency of our approach, we conduct fine-tuning experiments using the LLaMA2-7B-chat-hf model on the OpenBookQA dataset, with results presented in Fig.5. This setup allows us to compare the on-device memory footprint of different personalization methods under realistic fine-tuning conditions. As baseline methods such as STAR, ASK-LLM, and DEFT-UCS focus solely on representative sample selection from the unlabeled candidate pool D_u and perform full or standard PEFT-based adaptation (e.g., LoRA across all layers), their memory consumption during training remains constant and relatively high. These approaches do not incorporate any mechanism for selective model adaptation, leading to unnecessary overhead in both GPU memory and storage, even when only a subset of data is used for supervision.

In contrast, Flexora employs a gradient-based layer selection strategy to reduce the number of trainable layers, achieving the lowest memory footprint among the baselines. However, its selection criterion based on gradient-based layer selection. As a result, it fails to capture the dynamic distribution shifts that occur throughout the training process, leading to suboptimal adaptation. EcoTune, while incurring slightly higher memory usage than Flexora due to its dynamic layer importance tracking, achieves better downstream performance. This trade-off is well justified, as the marginal increase in memory enables the model to adapt more responsively to changing data distributions. Moreover, compared to full-adaptation baselines like STAR, EcoTune reduces memory consumption by 1.4 GB, demonstrating that selective adaptation can yield substantial resource savings without compromising accuracy. Overall, EcoTune strikes a balance between efficiency and performance, making it particularly suitable for deployment in edge devices where both high accuracy and low resource usage are critical.

In Appendix A.4, we provide a detailed analysis of the communication and latency costs associated with EcoTune’s iterative process in a real-world testbed. This evaluation highlights the efficiency of EcoTune in practical deployment scenarios, considering both network overhead and computational latency.

5.3 Trade-off Between Accuracy and Annotation Budget

To evaluate EcoTune’s ability to balance fine-tuning effectiveness and black-box annotation costs, we conduct experiments under varying annotation budgets, as shown in Fig. 6 and 7. This setting simulates real-world scenarios where cloud API usage is limited

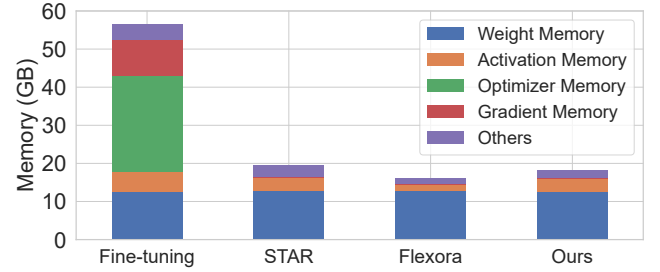


Figure 5: Memory breakdown of our method and various baselines using LLaMA2-7B-chat-hf on OpenBookQA dataset. Note that the batch size is set to 4.

by cost or latency constraints. The results demonstrate that EcoTune consistently outperforms all baseline methods across multiple datasets and budget levels, highlighting its robust generalization and efficient use of supervision signals. EcoTune’s superior data efficiency is attributed to its dual optimization strategy: selecting the most informative samples for cloud annotation and adapting only the critical model components. This approach enables EcoTune to achieve higher downstream accuracy under the same budget by prioritizing representative and semantically diverse samples, which accelerates learning and improves generalization.

Moreover, EcoTune significantly reduces cloud API expenditure, cutting annotation costs by up to 20%-60% compared to state-of-the-art baselines. This cost reduction is achieved without sacrificing performance, thanks to its intelligent sample selection mechanism that avoids redundant or low-value queries to the black-box LLM. Notably, on the ARC-Easy task, EcoTune achieves a strong accuracy of 76.48% while utilizing only 40% of the unlabeled data, a fraction of the annotation budget required by full-data methods. In contrast, DEFT-UCS achieves 76.89% when trained on the complete dataset. The near-parity in performance with less than half the labeled data highlights EcoTune’s exceptional sample efficiency. This result underscores its potential for practical deployment in resource-constrained environments, where minimizing cloud dependency is critical for scalability and user privacy.

5.4 Ablation Study

Effectiveness of data-efficient cloud annotation. To evaluate the effectiveness of our proposed MAB-based data selection strategy,

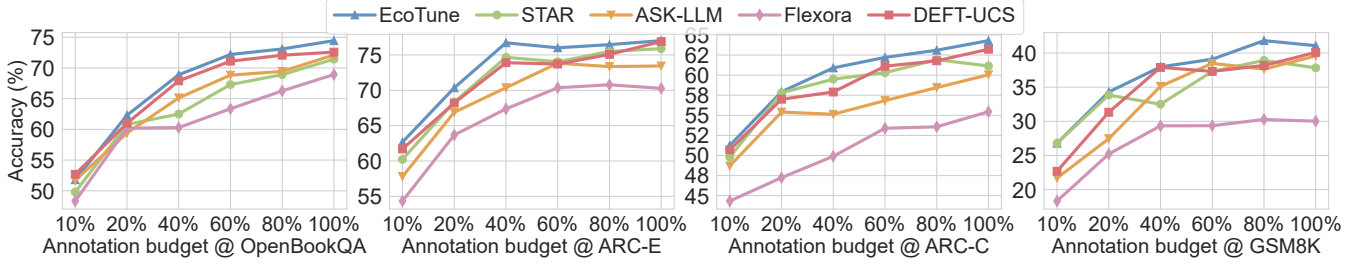


Figure 6: Accuracy–Cost trade-off of the LLaMA2-7B-chat-hf model across OBQA, ARC-E, ARC-C, and GSM8k benchmarks.

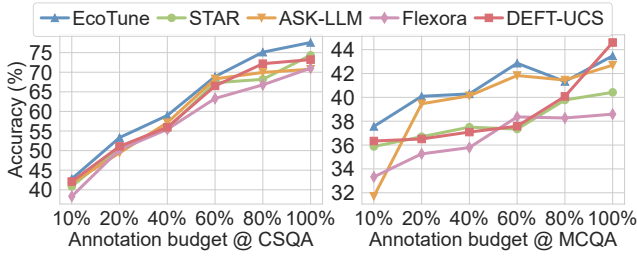


Figure 7: Accuracy–Cost trade-off of the LLaMA2-7B-chat-hf model across CSQA and MCQA benchmarks (continued).

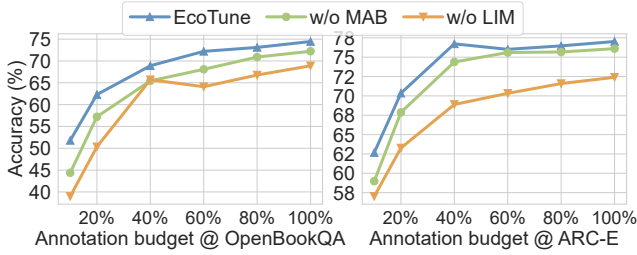


Figure 8: The impacts of data-efficient cloud annotation and Layer importance metric on the performance of OpenBookQA and ARC-E.

we conduct an ablation by removing this module and instead randomly selecting samples from the candidate pool (D_u) at each iteration on the OpenBookQA and ARC-E tasks. This baseline simulates a scenario where no intelligent prioritization is applied, and cloud annotation resources are allocated uniformly across interactions without considering their potential impact on model performance. As shown in Fig. 8, random selection consistently yields suboptimal performance across all annotation budget levels, resulting in an accuracy drop of 2.23%–7.41% on OpenBookQA and 0.92%–3.68% on ARC-E. These results highlight the critical role of intelligent data selection in the black-box cloud annotation pipeline, where each query incurs non-negligible cost and latency. In contrast to random sampling, EcoTune’s MAB-based strategy actively learns which types of interactions are most informative for personalization, balancing exploration and exploitation to identify samples that are both representative and semantically diverse. By focusing

annotation efforts on high-impact instances, EcoTune maximizes knowledge gain per query, enabling more effective and efficient model adaptation under strict resource constraints.

Effectiveness of layer importance metric. We further investigate the importance of selective model adaptation in fine-tuning by replacing our layer importance mechanism with a baseline that randomly selects layers of the SLM to inject LoRA adapters, while freezing the rest. As shown in the results, randomly selecting layers for adaptation leads to severe performance degradation. For instance, on the OpenBookQA task with a 10% annotation budget, this approach results in a 12.82% accuracy drop compared to EcoTune. This demonstrates that different layers play distinct functional roles during training, and randomly updating a subset of them disrupts the pre-trained model’s feature space and hampers effective knowledge transfer. In contrast, EcoTune’s layer importance metric ensures that only the critical layers are adapted, preserving model coherence while enabling efficient personalization.

6 Conclusion

In this work, we present EcoTune, a budget-aware framework for edge–cloud collaborative model adaptation that addresses the dual challenges of label scarcity and resource constraints in on-device personalization. By co-optimizing data-efficient cloud annotation and selective on-device model adaptation, EcoTune enables efficient LLM personalization under stringent annotation and computational budgets. Experiments show that EcoTune reduces annotation costs by up to 20%–60% while maintaining strong accuracy, outperforming existing baselines. Our work demonstrates the effectiveness of co-designing data selection and model adaptation for efficient, scalable on-device intelligence. In the future, we plan to extend EcoTune to support multi-device collaborative adaptation and more efficient personalization under heterogeneous device conditions, further improving its adaptability in dynamic mobile and edge environments.

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A Appendix

A.1 Embedding Model Selection

In this section, we present an additional experiment comparing the all-MiniLM-L6-v2 encoder [37] and the larger Qwen3-Embedding-0.6B encoder [56] in terms of both computational efficiency and performance on the OpenBookQA dataset. As shown in Table 2,

despite its compact size, the all-MiniLM-L6-v2 encoder demonstrates impressive performance across a variety of natural language understanding tasks, particularly in edge-device settings where computational and memory resources are constrained. Replacing the all-MiniLM-L6-v2 with the larger embedding model resulted in a significant increase in both feature extraction time (a 9.02× increase) and clustering time (a 1.938× increase), while the accuracy improvement for the small language model (SLM) was modest (+0.17%). Based on these experimental results, we conclude that the all-MiniLM-L6-v2 encoder strikes an optimal balance between accuracy and computational efficiency, especially in environments with limited computational resources.

Model	Param.	Mem.	Feature Extraction	Clustering	SLM Acc.
all-MiniLM-L6-v2	22.7M	87.7MB	2.2882±0.13s	0.4287±0.02s	74.45
Qwen3-Embedding-0.6B	0.6B	1.12GB	20.64±0.11s	0.8309±0.09s	74.62

Table 2: Comparison of different embedding model on OpenBookQA.

A.2 Cloud Model Selection for Annotation.

In this experiment, we evaluate the impact of different cloud annotators on the performance of the SLM under tight cloud annotation budgets. In our main experiment, we use the Qwen2.5-Instruct-14B as the default cloud annotator, as it provides high-quality pseudo-labels while maintaining competitive downstream SLM performance at a moderate cost. We further evaluate Qwen2.5-Instruct-32B, a higher-capacity cloud model, to examine whether stronger annotators lead to noticeable accuracy improvements. Compared to Qwen2.5-Instruct-14B, the 32B model incurs approximately twice the input/output token cost, with pricing increasing from 1 RMB/M (input) and 3 RMB/M (output) to 2 RMB/M (input) and 6 RMB/M (output). As shown in Table 3, we conduct experiments on six common benchmarks. While 32B achieved modest SLM accuracy gains (ranging from 0.35% to 3.2%) across these benchmarks, the improvement was relatively small compared to the increased annotation budgets of using a larger model. These results indicate that while stronger annotators can reduce label noise to some extent, the gains diminish as model capability increases. For edge scenarios where computational resources are constrained, moderately sized cloud-based model offer a favorable trade-off between label quality and computational efficiency.

Cloud Annotation	GSM8K	ARC-C	ARC-E	CSQA	OBQA	MCQA
14B	41.08	64.29	77.02	77.59	74.45	43.48
32B	43.22	65.79	80.22	78.98	74.80	45.13

Table 3: Comparison of SLM performance with different cloud annotators.

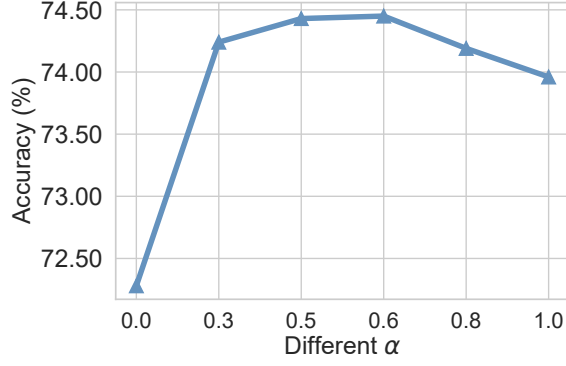


Figure 9: Performance on OpenBookQA with different α

A.3 Sensitivity Analysis of α

In this experiment, we evaluate the sensitivity of the hyperparameter α , which is used to balance the layer representational uniqueness and gradient contribution to loss reduction. The value of α influences how much the model emphasizes diverse, non-redundant layers versus focusing on gradients that contribute to the reduction of loss. We conducted the sensitivity evaluation on the OpenBookQA dataset using different values of $\alpha \in \{0.0, 0.3, 0.5, 0.6, 0.8, 1.0\}$. The SLM accuracy results for these settings are shown in Fig.9. The results show that both components, representational uniqueness and gradient contribution, are essential for the SLM’s performance. When α is set to 0.0 or 1.0, the performance degrades by more than 1.5%, indicating that the model performs poorly when it either entirely neglects one of the components or overly prioritizes one at the expense of the other. The optimal trade-off is observed at $\alpha = 0.6$, where the model achieves the highest accuracy and aligns with our goal of selecting diverse, non-redundant layers. Therefore, we set $\alpha = 0.6$ as default in our experiments.

A.4 Latency Analysis of EcoTune.

In this experiment, we provide a detailed analysis of the communication and latency costs associated with EcoTune’s iterative process in a real-world testbed. We evaluate the overhead introduced by EcoTune on the OpenBookQA dataset, using an RTX 3090 GPU and a 40% subset selection of the dataset. The overall latency of the process is expressed as follows:

$$T_{total} = T_e + T_c + \sum_n (T_{MAB} + T_f) + T_a \quad (11)$$

where T_e and T_c represent the feature extraction and clustering times, respectively, which are performed once before the training process. The parameter n denotes the number of data selection iterations required to meet the predefined budget. T_{MAB} and T_f refer to the times for the MAB process and local finetuning, respectively. Finally, T_a represents the annotation time for calling the cloud model.

Specifically, feature extraction takes 2.2882 ± 0.13 seconds, and clustering takes 0.4287 ± 0.02 seconds. After this initial setup, the subsequent rounds of the EcoTune process require only 14.8 ± 0.86 seconds for MAB selection, which is negligible compared to the total time required for local fine-tuning (48 minutes). This shows that EcoTune incurs minimal overhead once the initial feature extraction and clustering are completed. In terms of cloud annotations, all methods requiring cloud annotations incur similar per-query latency when operating over a campus wired network with a 1000 Mbps connection. To mitigate this latency, we employ batched, parallelized API calls with 64 concurrent requests. This approach significantly reduces the communication delay and helps optimize overall processing time. Furthermore, EcoTune offers a 20-60% reduction in the annotation budget, which contributes to a substantial reduction in total communication latency. This reduction in both annotation cost and communication latency demonstrates the efficiency of EcoTune, making it a viable solution for edge-computing scenarios with limited resources.